

AN ASYMPTOTIC FORMULA FOR THE r^{2q} ANHARMONIC BOUND STATES

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We consider the general polynomial potential $V(r) = \mu r^2 + \nu r^4 + \dots + \lambda r^{2q}$ and derive the leading-order $N \gg 1$ asymptotic formula for the harmonic-oscillator projections $\langle N|\psi \rangle$ of its normalizable wavefunctions.

The generalized anharmonic-oscillator equation

$$[-d^2/dr^2 + l(l+1)/r^2 + \mu r^2 + \lambda r^{2q}] \psi(r) = E\psi(r),$$

$$\lambda > 0, \quad l = -1, 0 \quad \text{or} \quad l = 0, 1, \dots, \quad (1)$$

is a real methodical laboratory in quantum mechanics [1]. Indeed, the anharmonicity r^{2q} with integer $q \geq 2$ is too singular for the simple-minded perturbation expansions and sometimes even for their re-summations [2]. Thus, it is difficult to treat the coupling λ as a small parameter in eq. (1).

The numerical determination of binding energies in the harmonic-oscillator "unperturbed" basis $|n\rangle$ is straightforward. The special $(2q+1)$ -diagonal matrix form of (1)

$$\sum_{n=m-q}^{m+q} H_{mn} \langle n|\psi \rangle = E \langle m|\psi \rangle, \quad m = 0, 1, \dots, \quad (2)$$

and an asymptotic growth of the matrix elements

$$H_{mn} = \lambda m^q \binom{2q}{m-n} + h_{mn}, \quad h_{mn} = O(m^{q-1}), \quad (3)$$

lead to a quick convergence of energies with respect to the truncation of (2) [3]. For the wavefunctions, an estimate

$$(-1)^n \langle n|\psi \rangle \approx \text{const}, \quad n \gg 1, \quad (4)$$

may also be obtained within the framework of the Feshbach projection operator method [4]. Its consequences are less pleasing: The small errors in H_{mn} may make a big difference in $\|\psi\|$ or in the various ex-

pectation values of the type $\langle \psi|0|\psi \rangle$. The explicit higher-order corrections to (4) are needed.

An analysis of the smooth functions $\xi_n = \langle n|\xi \rangle = (-1)^n \langle n|\psi \rangle$ represented by the Taylor approximants

$$\xi_{N+k} \approx \sum_{m=0}^M (k^m/m!) \xi_N^{(m)}, \quad N \gg 1, \quad M \gg 1, \quad (5)$$

with an assumption of the quickly decreasing derivatives

$$|\xi_N^{(k)}| \gg |\xi_N^{(k+1)}|, \quad k \gg k_0(N), \quad (6)$$

proved useful in the special cases $q=2$ [5] and $q=3$ [6]. Here, in accord with the conjecture of ref. [5], the same approximation procedure will be applied to the general $q \geq 2$ problem (1).

In the first step, we notice that the algebraic form of (2) provides an easy explanation for the asymptotic degeneracy (4) of the physical (normalisable) and unphysical (non-normalisable) asymptotics of $\langle n|\xi \rangle$, $n \rightarrow \infty$. Indeed, we may interpret r^2 asymptotically as a difference operator [7]

$$N^{-1} \langle N|r^2|\xi \rangle = -\xi_{N-1} + 2\xi_N - \xi_{N+1} + O(1/N), \quad (7)$$

which is asymptotically dominant in (2)

$$\langle N|r^{2q}|N' \rangle = O(N^q) \gg N \gg 1.$$

As a consequence of the second estimate

$$\begin{aligned} N^{-1} \langle N| -d^2/dr^2 + l(l+1)/r^2 |\xi \rangle \\ = \xi_{N-1} + 2\xi_N + \xi_{N+1} + O(1/N), \end{aligned} \quad (8)$$

the first $q - 1$ approximants reduce our Schrödinger equation (2) to the relations

$$[\lambda r^{2q} + O(N)] \sum_{n=N-q}^{N+q} |n\rangle (-1)^n \langle n|\xi\rangle = 0. \quad (9)$$

Of course, no quantization distinguishing between $\|\psi\| < \infty$ and $\|\psi\| = \infty$ is possible when we neglect the $O(N)$ kinetic energy: From the formal point of view, the difference operator r^{2q} reproduces merely the asymptotically constant solution (4). On physical grounds, we have to incorporate (8) in (9) on any level of approximation.

The separate treatment of the kinetic and potential parts of the hamiltonian H will be a physically motivated starting point of our present improvement of (4), with a mathematical consistency to be verified a posteriori. In the leading-order approximation, we may employ (3), (7) and (8) and replace (9) by the relation

$$\begin{aligned} \lambda \sum_{n=N-q}^{N+q} \left[\binom{2q}{N-n} + O(1/N) \right] (-1)^n \xi_n \\ + N^{1-q} \sum_{m=N-1}^{N+1} \left[\binom{2}{N-m} + O(1/N) \right] \xi_m = 0. \end{aligned} \quad (10)$$

For sufficiently large N and M , we may use also (5) and convert (10) (a difference equation of the order $2q$, [7]) into an approximate differential equation

$$\sum_{m=0}^M (1/m!) (\lambda c_m + N^{1-q} d_m) \xi_N^{(m)} = 0 \quad (11)$$

of the order M .

Proposition. When we choose $M = 2q$ and omit all the correction terms, we obtain the first non-trivial explicit form

$$(-1)^q \xi_N^{(2q)} + 4\lambda^{-1} N^{1-q} \xi_N = 0 \quad (12)$$

of the $N \gg 1$ asymptotic Schrödinger equation (11).

Proof. From a comparison of (10) and (11), we may see that $d_0 = 4$ and the coefficients c_k may be defined by the formula

$$\sum_{m=0}^{2q} x^{m-q} \binom{2q}{m} (m-q)^k = (x \, d/dx)^k x^{-q} (1+x)^{2q} \quad (13)$$

in the limit $x \rightarrow x_0 = -1$. A change of variables

$$x = e^{2t}, \quad x_0 = e^{2t_0}, \quad t_0 = \frac{1}{2} i\pi,$$

converts (13) into the function

$$2^{2q-k} (d^k/dt^k) (\cosh t)^{2q},$$

where each differentiation lowers the exponent of $\cosh t$. Since $\cosh t \rightarrow 0$ in the $t \rightarrow t_0$ limit, we get immediately $c_k = 0$ for all $k < 2q$. The first nonzero coefficient is easily evaluated, $c_{2q} = (-1)^q (2q)!$ and the proof is completed.

Let us solve now (12) in the leading-order approximation. In order to produce the factor N^{1-q} by the repeated differentiation, we put

$$\xi_N = \exp[\alpha(N) N^{1-\beta}/(1-\beta)], \quad 0 < \beta < 1, \quad (14)$$

and get $\xi_N^{(1)} = [\alpha N^{-\beta} + O(N^{-1})] \xi_N, \dots$, i.e.

$$\beta = (1/2q)(q-1). \quad (15)$$

In the asymptotically N -independent coefficients

$$\alpha(N) = \alpha = (4/\lambda)^{1/2q} e^{i\varphi} \quad (16)$$

the $2q$ phase factors $e^{i\varphi}$ are determined by the simple relation

$$e^{2iq\varphi} = (-1)^{q+1}. \quad (17)$$

Our construction is completed — we may verify now that our asymptotic solution (14)–(17) of recurrences (2) satisfies the requirement (6) with $k_0 = 0$ and is compatible also with the other assumptions made during its derivation.

From the physical point of view, we may conclude that

(i) Eq. (17) has $2q$ different solutions $\varphi = \varphi^{(\pm)}$,

$$\varphi^{(+)} = (s/q)\pi, \quad \varphi^{(-)} = \pi + (s/q)\pi,$$

$$s = m - \frac{1}{2}(q+1), \quad m = 1, 2, \dots, q. \quad (18)$$

Hence, the full degeneracy of asymptotics (4) is removed completely.

(ii) Of course, the solution of (12) is represented by a superposition of ξ 's over all the roots (18) in general. Up to one exception ($q = 2t + 1 = \text{odd}$ and $s = 0$ or $m = t + 1$), these solutions oscillate with $N \rightarrow \infty$. This is a peculiarity of the general anharmonic oscillator interaction (1).

(iii) For all the factors $\varphi = \varphi^{(+)}$, the real part of α (16) is positive. The exponentially growing wavefunctions (14) with $\varphi = \varphi^{(+)}$ should be discarded as unphysical, $\|\psi\| = \|\xi^{(+)}\| = \infty$.

(iv) The second set of phases $\varphi = \varphi^{(-)}$ with $\text{Re } \alpha < 0$ specifies q independent exponentially decreasing solutions ξ of the differential equation (12), i.e., the "Jost" [5] asymptotics of the Schrödinger difference equation (2). Each normalizable bound-state wavefunction will coincide with their superposition,

$$\langle N|\psi\rangle = (-1)^N \sum_{m=1}^q h_m \exp \left[-\rho_N \sin \left(\frac{2m-1}{2q} \pi \right) - i\rho_N \cos \left(\frac{2m-1}{2q} \pi \right) \right],$$

$$N \gg 1, \quad \rho_N = (4N/\lambda)^{1/2} q N^{1/2} / (1 - \beta) \gg 1, \quad (19)$$

in the asymptotic region. This is our main result.

(v) The leading-order improvement (19) of (4) generalizes the algebraic $q \leq 3$ formulae of refs. [5] and [6]. It remains valid also for the general polynomial potentials

$$V(r) = \mu r^2 + \nu r^4 + \dots + \sigma r^{2q-2} + \lambda r^{2q}, \quad (20)$$

since all the subdominant anharmonicities contribute to the error component in

$$\rho_N \sim N^{1-\beta} + O(N^\beta) \quad (21)$$

only. The same errors may be expected to appear also after an explicit inclusion of the corrections h_{mn} in (3). If needed, these corrections would incorporate an explicit dependence on $\sigma, \dots$ in (19) and may be obtained in analogy with $q = 2$ [5].

(vi) When we write the explicit error estimate (21) in (19),

$$\langle N|\psi\rangle = (-1)^N \exp[-\rho_N \sin(\pi/2q)] \times \{h_1 \exp[O(N^\beta)] + h_2 \exp(-\rho_N \gamma) + \dots\},$$

we have

$$\text{Re } \gamma = 2 \sin(\pi/q) \cos(\pi/q) > 0 \quad \text{for } q > 2$$

and see that the $m = 2, 3, \dots, q-1$ contributions are very small. Numerically, they disappear completely in the error background specified by the first and last terms. Hence, in contrast to $q = 2$, the general truncation-resembling symmetry [5] between the q -parametric "Jost" and "regular" solutions of recurrences (2) is broken. For $q \geq 3$, the two-parametric leading-order formulae of the type

$$\begin{aligned} \langle N|\psi\rangle &= (-1)^N \exp[\rho_N \sin(\pi/2q)] \\ &= h_1 \exp[-i\rho_N \cos(\pi/2q)] \\ &\quad + h_q \exp[i\rho_N \cos(\pi/2q)] \\ &= A \cos[\rho_N \cos(\pi/2q) + B], \quad N \gg 1, \end{aligned} \quad (22)$$

survive the inclusion of errors and may be used as asymptotics in the practical numerical computations, say, of the matrix elements $\langle \psi|O|\psi\rangle$.

(vii) As an example of a direct and specific numerical application of our algebraic analysis, we may recall, e.g., the recent "fixed-point" perturbation theory [8] where the eigenvalue calculations are based on a preliminary algebraic construction of the effective hamiltonians (cf. also ref. [9] in a somewhat related context). The simplest nontrivial ($q = 2$) numerical illustration of such a procedure may be found elsewhere [10].

(viii) In contrast to refs. [5,6], we approximate here separately the kinetic and potential energies [cf. (10)]. This is equivalent to an asymptotic modification of the potential

$$\begin{aligned} \langle N+m|r^{2q} + \dots |N+n\rangle \\ = [N/(N+m)]^{q/2} \langle N+m|r^{2q} + \dots |N+n\rangle \\ \times [N/(N+n)]^{q/2} [1 + O(1/N)], \quad N \gg 1. \end{aligned} \quad (23)$$

Mutatis mutandis, we may use the same trick for derivation of the higher-order corrections in a systematic way. Thus, the asymptotic and perturbation-type ($1/n \ll 1$) expansions of $\langle n|\psi\rangle$ might be obtained by our present algebraic technique in principle.

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